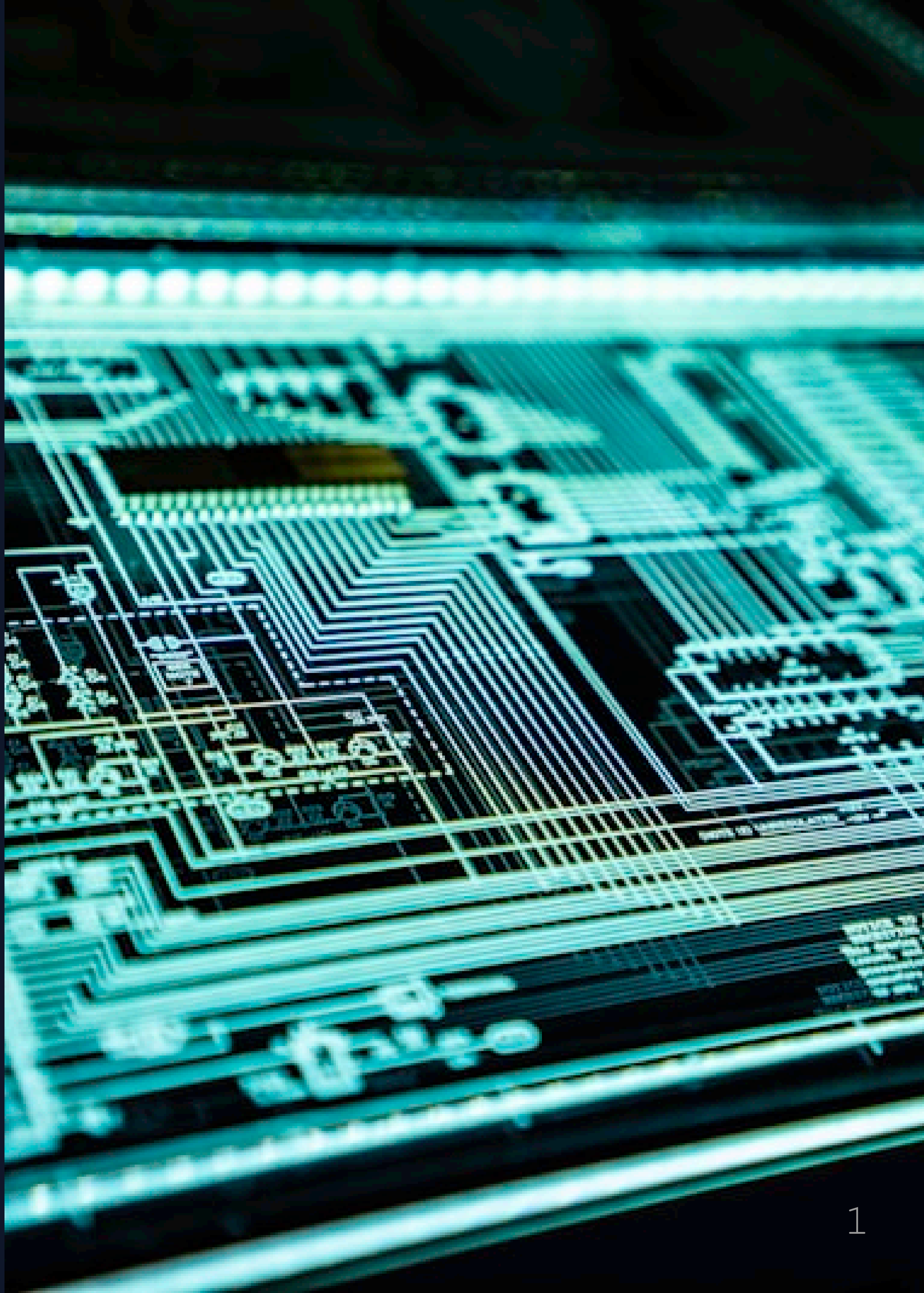
🌃 L01: Introduction to Python

Print() & Math Operators

Neon City Cyber Academy

Your coding journey begins here



👋 Meet Your Team

Emily Chen - *Your Handler*

> "Welcome to the Academy, recruit. I'll guide you through your training."

Cole Martinez - *Tech Lead*

> "I'll teach you the tools. Master them, and you'll be unstoppable."

🐍 What is Python?

A **programming language** - your tool to command computers.

Why Python?

- 🎮 **Games:** Minecraft mods, indie games
- 🤖 **AI:** ChatGPT, image generators
- 🌐 **Web:** Instagram, YouTube, Spotify
- 🔐 **Security:** Ethical hacking tools

Used by: Google, Netflix, NASA, Tesla

Emily Says:

> "Python is named after Monty Python, not the snake! The creators had a sense of humor."

> "Every app you use daily - TikTok, Discord, YouTube - has Python code inside it."

Python = Easy to learn + Incredibly powerful

📢 Your First Tool: print()

The `print()` function displays text on screen.

```
print("Hello, World!")
```

Breakdown:

- `print` → The command
- `()` → Parentheses (required!)
- `"..."` → Text in quotes (a "string")

Think of it as your computer's megaphone!

🎯 Challenge #1: Your First Program

Type this EXACTLY:

```
print("Hello, Neon City!")
```

What happens:

1. Python reads the command
2. Sees `print()`
3. Grabs the text
4. Displays it instantly ⚡

🔄 Sequential Execution

Python runs code **top to bottom,** one line at a time.

```python
print("Starting mission...")
print("Loading systems...")
print("Ready!")
```

Output:

```
Starting mission...
Loading systems...
Ready!
```

Each `print()` creates a **new line**.

💬 Cole Says:

> "Computers are dumb. They do EXACTLY what you tell them, in EXACTLY the order you say it."

> "If your code doesn't work, 99% of the time it's a typo. Check capitalization, quotes, and parentheses!"

Python is case-sensitive: `print` ≠ `Print`

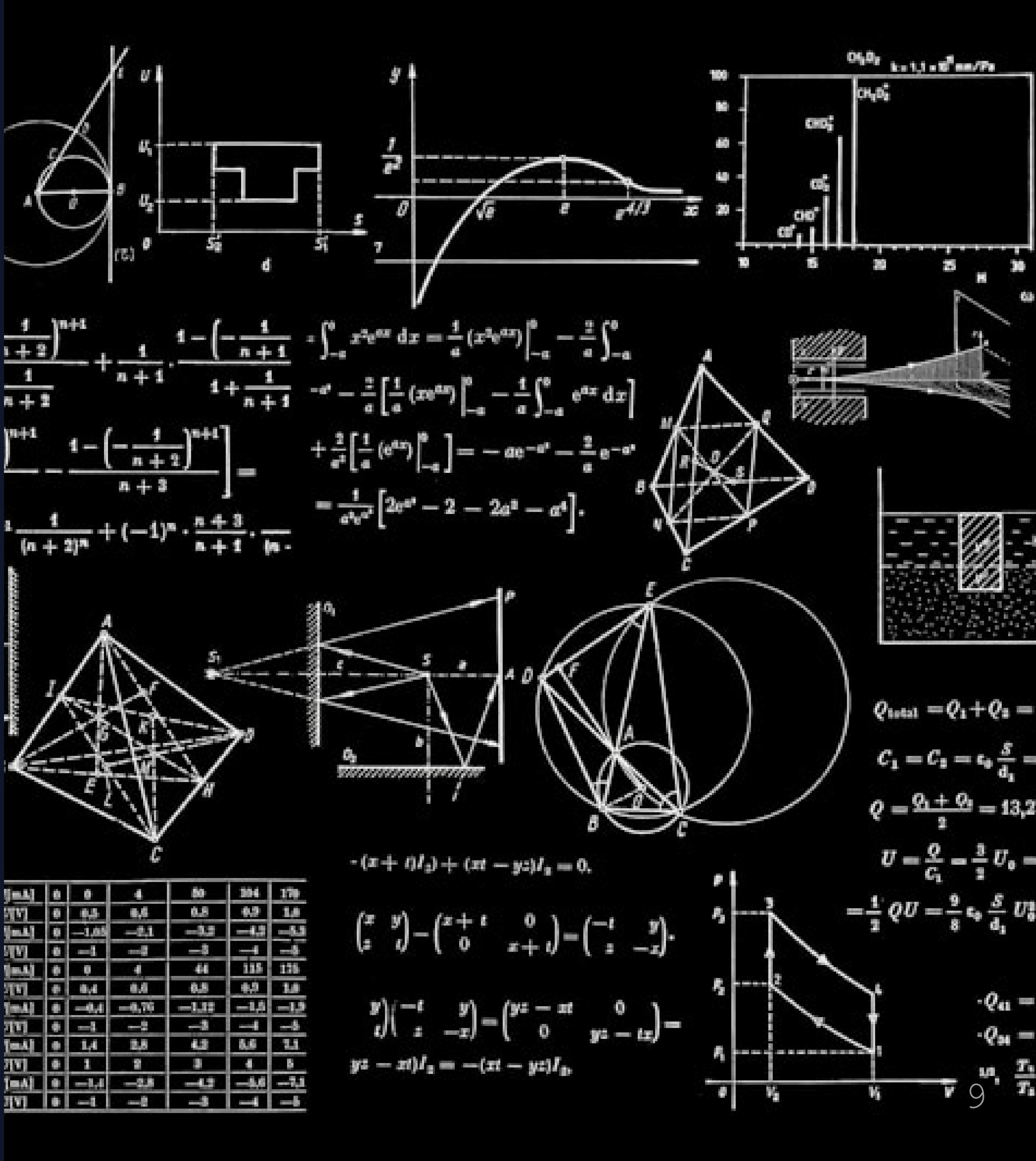
🧮 Python as a Calculator

You can print **calculations:**

```
print(5 + 3)        # Output: 8
print(10 - 4)       # Output: 6
```

Python solves the math FIRST, then prints the result.

Your computer = Super calculator



+ Operator #1: Addition

```
print(5 + 3)        # 8
print(100 + 27)     # 127
print(1 + 1 + 1)    # 3
```

Uses:

- Game scores
- Shopping carts
- Counting items

Pro tip: Use spaces for readability

`5 + 3` is better than `5+3`

— Operator #2: Subtraction

```
print(10 - 3)       # 7
print(100 - 25)     # 75
print(5 - 5)        # 0
print(3 - 10)       # -7 (negative!)
```

Uses:

- Reducing health
- Countdown timers
- Calculating change

✖ Operator #3: Multiplication

```
print(5 * 3)        # 15
print(7 * 7)        # 49
print(2 * 10)       # 20
```

Cool trick:

```
print("=" * 10)    # ==========
```

Repeats text! We'll use this for borders.

÷ Operator #4: Division

```
print(10 / 2)        # 5.0
print(15 / 4)        # 3.75
print(7 / 3)         # 2.333...
```

Notice: Python always gives decimals with `/`

Even `10 / 5` returns `2.0` (not just `2`)

💬 Emily Explains:

> "Division always gives you a float (decimal number) because Python wants to be precise."

> "If you divide 10 cookies among 3 people, the answer isn't just '3' - it's '3.33 cookies each'. Python shows the exact math!"

⬇️ Operator #5: Floor Division

The `//` operator rounds DOWN to the nearest whole number.

```
print(10 // 3)     # 3 (not 3.33)
print(15 // 4)     # 3 (not 3.75)
print(7 // 2)      # 3 (not 3.5)
```

Real example:

"10 cookies, 3 people. How many does each get?"

`10 // 3 = 3` (with some left over!)

📐 Operator #6: Modulo (%)

Gets the **REMAINDER** after division.

```
print(10 % 3)       # 1 (10 ÷ 3 = 3 remainder 1)
print(15 % 4)       # 3
print(8 % 2)        # 0 (no remainder)
```

Uses:

- Check if number is even: `x % 2 == 0`
- Cycle through lists
- "What's left over?"

💬 Cole's Pro Tip:

> "Use `//` and `%` together:"

```
print(17 // 5)     # 3 (full groups)
print(17 % 5)      # 2 (leftover)
```

> "17 items, groups of 5: You get **3 complete groups** with **2 left over**!"

⚡ Operator #7: Exponent (**)

Raises a number to a power.

```
print(2 ** 3)       # 8 (2×2×2)
print(5 ** 2)       # 25 (5×5)
print(10 ** 3)      # 1000
print(2 ** 10)      # 1024 (kilobyte!)
```

This is how computers handle HUGE numbers!

Powers of 2 are everywhere in computing.

All 7 Operators

Symbol	Name	Example	Result
`+`	Addition	`5 + 3`	8
`-`	Subtraction	`10 - 4`	6
`*`	Multiplication	`6 * 7`	42
`/`	Division	`15 / 4`	3.75
`//`	Floor Division	`15 // 4`	3
`%`	Modulo	`15 % 4`	3
`**`	Exponent	`2 ** 5`	32

🔢 Order of Operations (PEMDAS)

```python
print(2 + 3 * 4)        # 14 (not 20!)
# 3*4=12 first, then +2

print((2 + 3) * 4)      # 20
# Parentheses first: 2+3=5, then *4
```

Parentheses = Your control over math order!

💬 Emily's Warning:

> "Order matters! This is the #1 cause of math bugs:"

```
print(100 - 20 * 2)    # 60
print((100 - 20) * 2)  # 160
```

> "Always use parentheses when you're unsure. It makes your code clearer!"

🎨 ASCII Art with Math

Combine print() and math for visual art!

```
print("=" * 30)
print("    CYBER ACADEMY")
print("=" * 30)
print("Level:", 5 + 3)
print("Points:", 100 * 10)
print("=" * 30)
```

Creativity + Code = Awesome!

🏆 Final Challenge: Build a Calculator

Create a program showing ALL 7 operations:

```
print("=" * 40)
print("     PYTHON CALCULATOR")
print("=" * 40)
print("25 + 7 =", 25 + 7)
print("25 - 7 =", 25 - 7)
print("25 * 7 =", 25 * 7)
print("25 / 7 =", 25 / 7)
print("25 // 7 =", 25 // 7)
print("25 % 7 =", 25 % 7)
print("25 ** 2 =", 25 ** 2)
print("=" * 40)
```

🐞 Common Errors

Error #1: Syntax Error

```
print(5 +)          # Missing number!
```

Error #2: Dividing by Zero

```
print(10 / 0)       # CRASH!
```

Error #3: Forgetting Quotes

```
print(Hello)        # Python looks for variable
```

Errors are normal! Read the messages - they help you.

💬 Cole's Final Words:

> "Professional developers see 100+ errors per day. We just know how to fix them fast."

> "Every expert was once a beginner. You took the first step today. Keep coding!"

✅ Lesson Complete!

You Mastered:

- ✅ Using `print()` to display output
- ✅ Sequential code execution
- ✅ **All 7 math operators**
- ✅ Order of operations (PEMDAS)
- ✅ ASCII art with code

Next Mission: L02 - Variables

Learn to STORE data, not just display it!

🚀 Your Assignment

Create **3 calculator programs** with:

1. Different numbers
2. Creative borders
3. Your own style

Practice makes perfect!

See you next lesson, Agent. 🌆